WEEK 3 · 23–26 NOVEMBER 2026

Validation methods and accuracy assessment

How good is your map, really — and how do you defend that number

Week 3 · Session 3 · practical: 21_validation_accuracy_assessment.ipynb

What you will be able to do

- Build and read an error matrix, and compute user's, producer's, F1 and IoU correctly
- Explain why kappa is no longer recommended, and what to report instead
- Design a train/test split that is *spatially* independent
- Draw a probability sample and estimate area-adjusted accuracy with confidence intervals
- Validate continuous products with bias, RMSE and Bland–Altman, not R^2 alone

1. What validation is for

A map is a model, so it has errors — the question is how many

THE STANDARD TO MEET

"It looks right on the map" is not a validation.

A validation estimates how many errors, of which kind, and where — for the whole map, not just the pixels you happened to label. And it comes with an uncertainty.

Three parts of a rigorous assessment

- **Sampling design** — how reference locations are chosen (random, stratified, systematic)
- **Response design** — how the true class at each location is determined, independently of the map
- **Analysis** — the estimators that turn the sample into accuracy and area, with confidence intervals
  - Skip any one of the three and the number you report is not defensible

Where reference data comes from

Source	Quality	Cost	Watch out for
Field survey / GPS points	Highest	Highest	Small sample, access bias, date mismatch
VHR image interpretation	High	Medium	Interpreter bias; needs a protocol
Existing maps (WorldCover, CLC)	Medium	Lowest	Its own errors, different legend and date
Crowd-sourced (Geo-Wiki, OSM)	Variable	Low	Uneven coverage and quality

Reference data must be independent

- Independent of the **training** pixels — otherwise you measure memorisation, not generalisation
- Independent of the **map**: never look at the map label while assigning the reference class
- Matched in **time**: WorldCover 2021 against a 2025 image contains real change, not only error
- Matched in **legend**: harmonise classes before comparing, and say how you did it

2. The error matrix

Everything else is derived from this table

Orientation and notation

- Convention in remote sensing: **rows = map class**, **columns = reference class**
- scikit-learn's `confusion_matrix(y_true, y_pred)` is the transpose — check before you read it
- Diagonal = agreement; off-diagonal = the two kinds of error
- **Commission** (row): mapped as X, is not X. **Omission** (column): is X, mapped as something else

DEFINITIONS

The metrics, and what each one answers

Metric	Formula	Question it answers
Overall accuracy	Σ diagonal / N	What fraction of the map is right?
User's accuracy (precision)	n_ii / row total	If the map says X, how often is it X?
Producer's accuracy (recall)	n_ii / column total	If it is X, how often did the map find it?
F1	2·UA·PA / (UA + PA)	One number balancing both, per class
IoU (Jaccard)	TP / (TP + FP + FN)	Overlap of mapped and true extent

PURE-PIXEL TEST SET

Notebook 21: a five-class Tanager map of Ho Chi Minh City

Class	UA	PA	F1	IoU
Tree cover	0.828	0.696	0.757	0.608
Grassland	0.509	0.657	0.573	0.402
Cropland	0.591	0.632	0.611	0.440
Built-up & bare	0.898	0.901	0.899	0.817
Water	0.806	0.859	0.832	0.712
Overall accuracy 0.763 · macro F1 0.734 · weighted F1 0.766				

Macro or weighted?

- **Macro** averages the per-class scores equally — small classes count as much as large ones
- **Weighted** averages by class frequency — dominated by whatever covers most of the scene
- Report both when classes are imbalanced; they differ here (0.734 vs 0.766)
- For a rare target class (a plume, a mangrove fringe), macro F1 or that class's F1 is the honest headline

Kappa: why it is no longer recommended

- κ = (p_o − p_e)/(1 − p_e) 'corrects' accuracy for agreement expected by chance
- The chance baseline is not meaningful for maps, and κ is highly correlated with overall accuracy
- It adds no information the error matrix does not already give (Pontius & Millones 2011; Foody 2020)
- **Instead:** report OA with a confidence interval, per-class UA/PA, and quantity vs allocation disagreement
  - Notebook 21: κ = 0.693, while Q = 0.051 and A = 0.186 — allocation is the real problem here

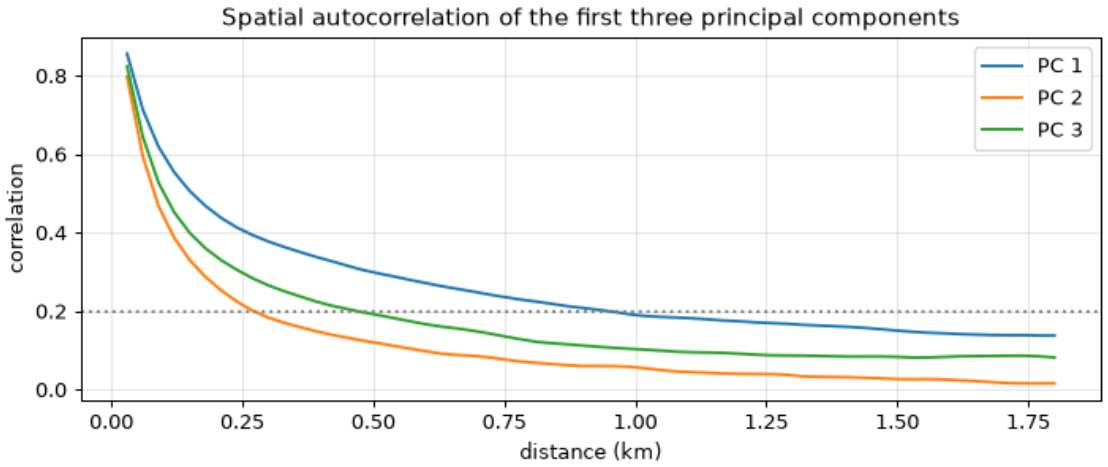
3. Independence

The spatial trap that inflates every accuracy figure

MEASURED IN NOTEBOOK 21

Random splits leak through space

- Neighbouring pixels are nearly identical — a random split puts them in both sets
- Random 5-fold CV: **OA 0.812 ± 0.011**
- Blocked CV, 100-px (3 km) blocks: **OA 0.764 ± 0.042**
- The gap is the optimism; the blocked number is what to expect in a new area



Accuracy falls as test blocks move away from training data

How to split properly

- Split by **blocks, sites or dates** — never by individual pixels
- Add a **buffer** so patch-based models cannot see test pixels inside their training patches
- Block size should exceed the spatial autocorrelation range of the target
- Report which design you used; 'random 80/20' on pixels is a red flag to any reviewer

4. Design-based accuracy and area

Olofsson et al. (2014) good practice

Why the test set above is still optimistic

- It contains only **pure** pixels — the easy ones, with mixed boundary pixels excluded
- Its class proportions follow the reference map, not the map being assessed
- It cannot give you **area** estimates, because it is not a probability sample of the map
- **Fix:** draw a random sample *from the map*, stratified by mapped class

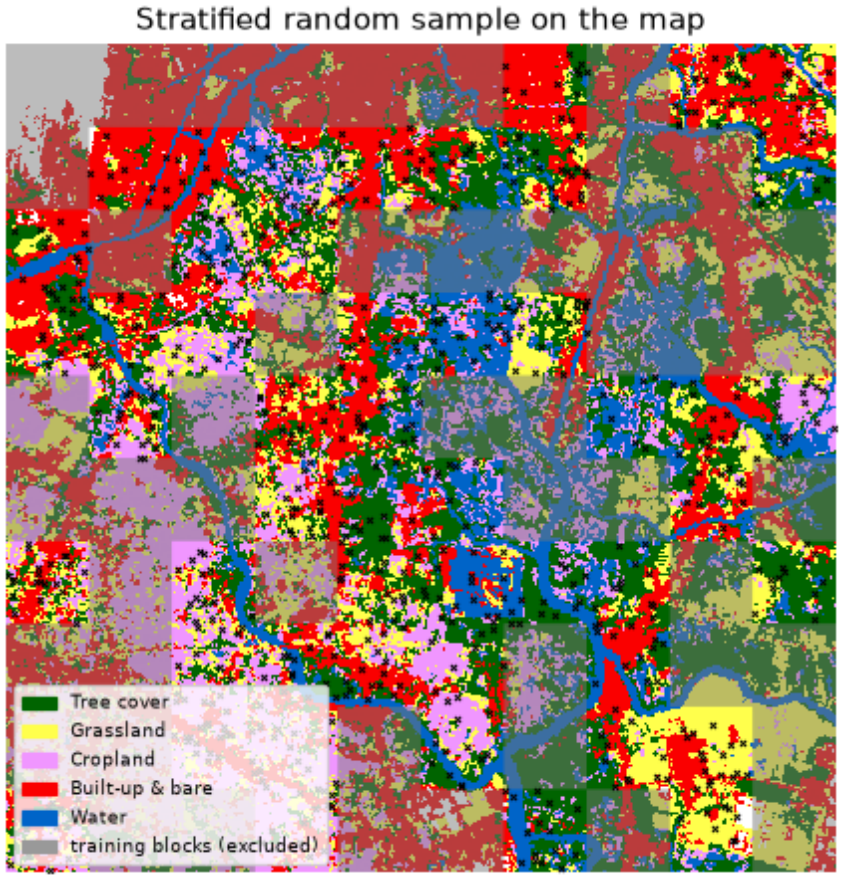
How many reference points do you need?

```
# Olofsson et al. (2014), Eq. 13 — target a standard error on overall accuracy
S_O = 0.015                                  # desired SE of OA (±0.03 at 95 %)
W   = mapped_area_fraction                   # per mapped class (strata weights)
U   = conjectured_user_accuracy              # a prior guess per class, e.g. 0.6-0.95

n = (np.sum(W * np.sqrt(U * (1 - U))) / S_O) ** 2     # total sample size
n_i = allocate(n, W, floor=50)                        # spread over strata, floor for rare classes
```

Notebook 21: n = 677 points over five strata, with a 50-point floor per class

Where the sample points land



677 stratified random points drawn from the map's own classes

From sample counts to area-weighted estimates

- Each stratum was sampled at a different rate, so counts must be reweighted by stratum area
- Estimated area proportion: **$\hat{p}_{ij} = W_i \cdot n_{ij} / n_{i+}$**
- Overall accuracy $\hat{O} = \Sigma\, \hat{p}_{jj}$; user's $\hat{U}_i = \hat{p}_{ii} / \hat{p}_{i+}$; producer's $\hat{P}_j = \hat{p}_{jj} / \hat{p}_{+j}$
- Error-adjusted area of class j = total area × $\hat{p}_{+j}$, with a 95 % confidence interval
  - Pixel counting ignores commission and omission, so it is biased — this corrects it

NOTEBOOK 21, 677 SAMPLE POINTS

Mapped area vs error-adjusted area

Class	Mapped (ha)	Adjusted (ha)	95 % CI	UA ± CI
Tree cover	1 897	2 320	± 223	0.687 ± 0.068
Grassland	1 150	974	± 173	0.394 ± 0.092
Cropland	1 306	1 250	± 195	0.407 ± 0.087
Built-up & bare	1 900	1 804	± 173	0.737 ± 0.065
Water	921	826	± 135	0.632 ± 0.102

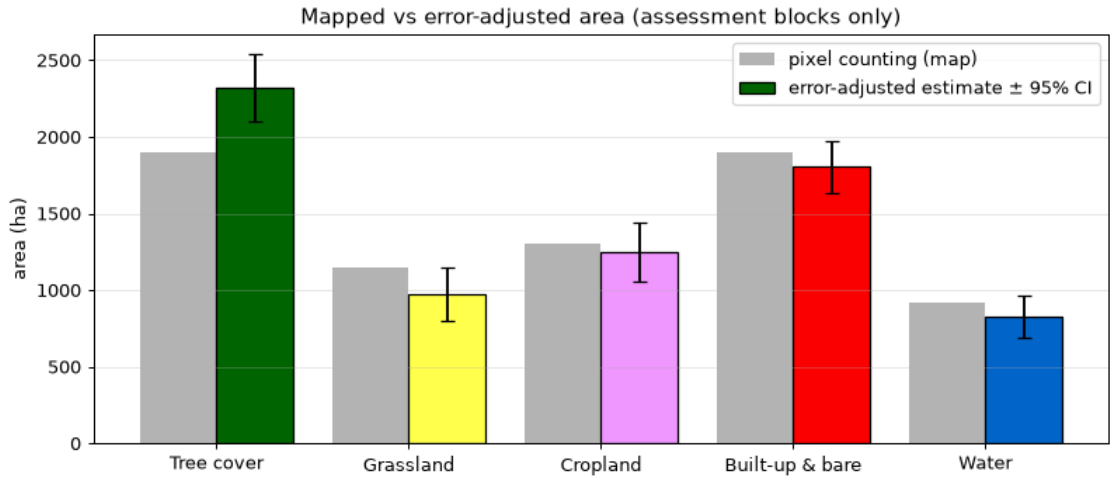
THE NUMBER THAT SURVIVES SCRUTINY

Pure-pixel test set: OA 0.763.
Probability sample of the same map: OA 0.595 ± 0.035.

Same map, same day, two evaluation designs. The lower number is the honest one — it includes the mixed pixels at every boundary, which is where maps really fail.

Accuracy depends on how mixed the pixel is

- Agreement by WorldCover purity of the 30 m pixel:
  - purity 1–5 (very mixed): **0.337**
  - purity 6–8: **0.462**
  - purity 9 (pure): **0.736**
- At 30 m, most error lives at class boundaries, not in field interiors



Area-weighted accuracy and area estimates with confidence intervals

Does the confidence interval actually work?

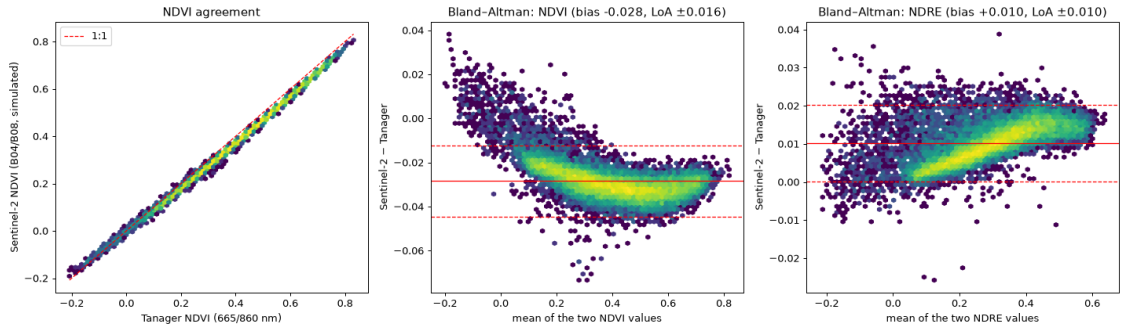
- Notebook 21 simulates 1 000 repeated samples from a known population
- Stratified: mean OA estimate 0.636, **95 % CI covers the truth 95.3 %** of the time
- Simple random: 0.638, coverage 94.3 % — also unbiased for OA
- The gain from stratification shows up for the **rarest class**: tighter UA intervals with the same budget

5. Continuous products

When the output is a number, not a class

Agreement metrics for continuous maps

- **Bias** — systematic offset (the thing R^2 cannot see)
- **MAE / RMSE** — typical error magnitude, RMSE punishes outliers
- **R^2 about the 1:1 line** — not Pearson r^2, which ignores offset and slope
- **Bland–Altman** — difference vs mean, reveals magnitude-dependent error



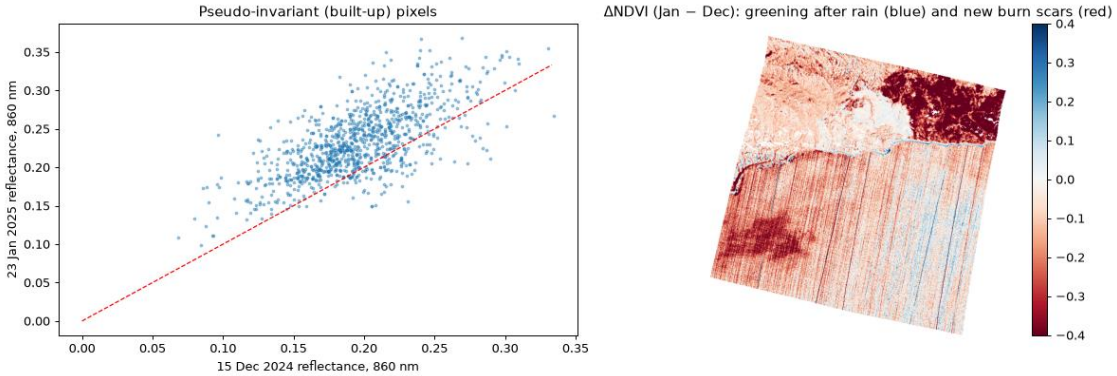
Simulated Sentinel-2 vs Tanager indices over 20 000 land pixels

The same comparison, four ways

Index	Bias	RMSE	R^2 (1:1)	Pearson r^2
NDVI, S2-simulated vs Tanager	−0.0285	0.0296	0.973	0.999
NDRE, S2-simulated vs Tanager	+0.0102	0.0114	0.994	0.999

Cross-date consistency on stable targets

- Compare two acquisitions over pixels that should not have changed
- 1 011 pseudo-invariant pixels between the two Malibu scenes
- Relative RMSE **25–36 %** depending on wavelength
- **Use:** a change smaller than that cannot be distinguished from noise



Pseudo-invariant targets, December 2024 vs January 2025

6. What to report

Turning this into marks and into trust

Minimum validation reporting for the project

Report	Because
Sampling and response design, in words	Shows the number is defensible
Error matrix (counts) + per-class UA/PA/F1	One OA hides class failures
OA with a 95 % confidence interval	A number without uncertainty is not a result
Area-adjusted area if you quote hectares	Pixel counting is biased
Random vs spatial split comparison	Proves you controlled for leakage
A baseline (threshold or index)	Shows what the complex model added
Three failure cases, with the spectra	Turns errors into understanding

PRACTICAL

Over to you

▶ `21_validation_accuracy_assessment.ipynb` · `100 min`

- Build the error matrix and compute UA, PA, F1 and IoU by hand, then check against scikit-learn
- Compare random and blocked cross-validation, and plot accuracy against block size
- Draw a stratified sample, then estimate area-adjusted accuracy and area with 95 % CIs
- Validate a continuous product with bias, RMSE and a Bland–Altman plot

KEY TAKEAWAYS

Recap

- Sampling design, response design and analysis — all three, or the number means little
- Report the **error matrix and per-class** accuracies, not a single overall figure
- Kappa adds nothing; quantity and allocation disagreement say more
- **Spatially** independent splits, always; random pixel splits inflate accuracy
- Quote **area-adjusted** areas with confidence intervals, not pixel counts
- For continuous products, R^2 without bias and RMSE is a half-truth

Next: Project build time this afternoon, then submission and presentations tomorrow

References and further reading

Olofsson, P. et al. (2014). Good practices for estimating area and assessing accuracy of land change. RSE 148, 42–57.

Stehman, S.V. & Foody, G.M. (2019). Key issues in rigorous accuracy assessment of land cover products. RSE 231, 111199.

Pontius, R.G. & Millones, M. (2011). Death to Kappa. International Journal of Remote Sensing 32, 4407–4429.

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Bland, J.M. & Altman, D.G. (1986). Statistical methods for assessing agreement. The Lancet 327, 307–310.

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